

## §1.2

# Types of Data

Who we study, what we measure, and what kind of thing we get back

# Today's Goal

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After completing this section, you should be able to:

- Define and distinguish between populations and samples
- Distinguish between parameters and statistics
- Classify data as qualitative or quantitative
- Classify quantitative data as discrete or continuous

# What You Will Be Graded On

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## LT.1 Types of Data

I understand the differences between various types of data (e.g., qualitative, quantitative, discrete, continuous, etc.). I can use this understanding to identify what type of data a given data set is.

# Who Are We Talking About?

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**A report says 3 in 10 Fresno State students work full time. It was written after asking 200 students. Which group is the report about?**

All of them — but only 200 were measured. Each group gets its own name.

# Observations and Their Values

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## Definition 1.2.1

**Data** are collections of observations. More precisely, data refers to the values associated with these observations. *Data* is plural; *datum* is singular.

So “the data *are*,” not “the data *is*.”



# The Whole Group

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## Definition 1.2.2

A **population** is the complete collection of all subjects being considered in a statistical study.

It is also the group your conclusions are meant to apply to.

A population need not be people. It can be batteries, fish, or storms.

# Everyone, or Only Some?

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## Definition 1.2.3

A **census** is the collection of data from every member of the population.

## Definition 1.2.4

A **sample** is a subcollection of members selected from a population.

# How the Sample Gets Chosen

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## Definition 1.2.5

**Sampling** is a process by which a sample is selected from a population.

The sample is the *result*. Sampling is the *action*.

Specific sampling methods come in §1.4.



# The Property You Record

## Definition 1.2.6

In statistics, a **variable** is a characteristic or property that can be determined for each member of the population. It is usually represented by a capital letter such as  $X$  or  $Y$ .

A person's height, a person's ethnicity, or whether or not a car passes a smog check.

# Which Number Can We Get?

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**The average height of all 25,000 students at a university, or the average height of 100 of them. Which can you actually compute?**

Only the second. The first is what we *want*; the second is what we can *get*.

# A Number From Each Group

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## Definition 1.2.7

A **statistic** is a numerical property of a sample.

## Definition 1.2.8

A **parameter** is a numerical property of a population.

# Telling the Two Numbers Apart

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A company wants to know the average commute time of all of its employees.

**Parameter.** The average over every employee. It describes the whole population, and it is *estimated*.

**Statistic.** Measuring everyone is impractical, so we take 60 employees. Their average describes the sample, and it is *calculated*.

# Finding the Parts of a Study

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A campus research office surveys 2,600 California community college students. Of those surveyed, 1,144 report working more than 20 hours a week.

The office uses this to estimate that 44% of California community college students work more than 20 hours a week.

Name the **population**, **sample**, **variable**, **statistic**, and **parameter**.



# The Five Parts

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Recall: 1,144 of 2,600 students surveyed work more than 20 hours a week.

- **Population.** All California community college students.
- **Sample.** The 2,600 students surveyed. Careful: 2,600 is the sample *size*, not the sample.
- **Variable.** Whether or not the student works more than 20 hours a week.
- **Statistic.**  $\frac{1144}{2600} = 0.44$ , or 44% — *calculated*.
- **Parameter.** 44% of *all* such students — *estimated*.

# Now You Try: Name the Parts

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**You Try 1.2.1.** A clinic checks 350 of the dogs it treats and finds 91 overweight.

1. Population? **All dogs the clinic treats.**
2. Sample? **The 350 dogs checked.**
3. Variable? **Whether the dog is overweight.**
4. Statistic?  $\frac{91}{350} = 26\%$ , **calculated.**
5. Parameter? **The percentage overweight among all of them.**

# Is That Number a Number?

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**Your student ID is written with digits. Does adding two student ID numbers tell you anything?**

No. Digits alone do not make data numerical. What matters is whether the values *count* or *measure* something, or merely *name* it.



# When the Numbers Mean Amounts

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## Definition 1.2.11

**Quantitative data** (also called **numerical data**) consists of numbers representing counts or measurements.

When reporting quantitative data, always include appropriate units.

# When the Values Are Names

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## Definition 1.2.12

**Qualitative data** (also called **categorical data**) consists of names or labels that represent categories.

A label may still be written with digits. That does not make it a measurement.

# Sorting Numbers From Labels

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1. The number of text messages a student sends in a day. **Quantitative** — it is a count.
2. The brand of phone a student uses. **Qualitative** — it names a category.
3. The ZIP code of a student's home address. **Qualitative**. Be careful here. It is written with digits, but it labels a place. Adding two ZIP codes produces nothing useful.

# Now You Try: Numbers or Names?

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**You Try 1.2.2.** For each variable, decide whether the data it produces is quantitative or qualitative.

1. The price of a textbook, in dollars. **Quantitative.**
2. A patient's blood type. **Qualitative.**
3. A runner's bib number in a race. **Qualitative.**
4. The number of pets in a household. **Quantitative.**

The test: would arithmetic on the values mean anything?

# Values You Can Count

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## Definition 1.2.14

**Discrete data** is quantitative data that can only take on a finite or countable number of values.

A rule of thumb: data that comes from *counting* is usually discrete.



# Values You Can Measure

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## Definition 1.2.15

**Continuous data** is quantitative data that can take on any value within a range, including fractional values.

And data that comes from *measuring* is usually continuous.

# Sorting Counts From Measurements

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1. The number of eggs a hen lays in a month. **Discrete** — a count. You cannot have 4.3 eggs.
2. The weight of a bag of oranges. **Continuous** — it can be measured to any precision.
3. The temperature of a cup of coffee as it cools. **Continuous** — it passes through every value in between.

# Now You Try: Counts or Measurements?

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**You Try 1.2.3.** For each variable, decide whether the data it produces is discrete or continuous.

1. The number of songs on a playlist. **Discrete.**
2. The rain that falls in a storm, in inches. **Continuous.**
3. The age of a tree, recorded to the nearest year. **Continuous** — the recording is rounded, the age is not.
4. The number of students absent from class. **Discrete.**



# Putting It Together

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